

Theorem 1.4.3. (Mader 1972)

[7.2.3]
[11.2.3]

Let $0 \neq k \in \mathbb{N}$. Every graph G with $d(G) \geq 4k$ has a k -connected subgraph. In fact, every such G has a $(k+1)$ -connected subgraph H such that $d(H) > d(G) - 2k \geq 2k$.

Proof. Put $\gamma := \varepsilon(G) (\geq 2k)$, and consider the subgraphs $G' \subseteq G$ such that

(1.2.2)
(1.3.1)
 γ

$$|G'| \geq 2k \quad \text{and} \quad \|G'\| > \gamma(|G'| - k). \quad (*)$$

Such graphs G' exist since G is one; let H be one of smallest order.

H

No graph G' as in $(*)$ can have order exactly $2k$, since this would imply that $\|G'\| > \gamma k \geq 2k^2 > \binom{|G'|}{2}$. The minimality of H therefore implies that $\delta(H) > \gamma$: otherwise we could delete a vertex of degree at most γ and obtain a graph $G' \subsetneq H$ still satisfying $(*)$. In particular, we have $|H| \geq \gamma$. Dividing the inequality of $\|H\| > \gamma|H| - \gamma k$ from $(*)$ by $|H|$ therefore yields $\varepsilon(H) > \gamma - k$, as desired.

It remains to show that H is $(k+1)$ -connected. If not, then H has a proper separation $\{U_1, U_2\}$ of order at most k ; put $H[U_i] =: H_i$. Since any vertex $v \in U_1 \setminus U_2$ has all its $d(v) \geq \delta(H) > \gamma$ neighbours from H in H_1 , we have $|H_1| \geq \gamma \geq 2k$. Similarly, $|H_2| \geq 2k$. As by the minimality of H neither H_1 nor H_2 satisfies $(*)$, we thus have

H_1, H_2

$$\|H_i\| \leq \gamma(|H_i| - k)$$

for $i = 1, 2$. But then

$$\begin{aligned} \|H\| &\leq \|H_1\| + \|H_2\| \\ &\leq \gamma(|H_1| + |H_2| - 2k) \\ &\leq \gamma(|H| - k) \quad (\text{as } |H_1 \cap H_2| \leq k), \end{aligned}$$

which contradicts $(*)$ for H . $\square$

1.5 Trees and forests

An *acyclic* graph, one not containing any cycles, is called a *forest*. A connected forest is called a *tree*. (Thus, a forest is a graph whose components are trees.) The vertices of degree 1 in a tree are its *leaves*,⁵ the others are its *inner vertices*. Every non-trivial tree has a leaf – consider, for example, the ends of a longest path. This little fact often comes in handy, especially in induction proofs about trees: if we remove a leaf from a tree, what remains is still a tree.

forest
tree
leaf

⁵ ... except that the *root* of a tree (see below) is never called a leaf, even if it has degree 1.

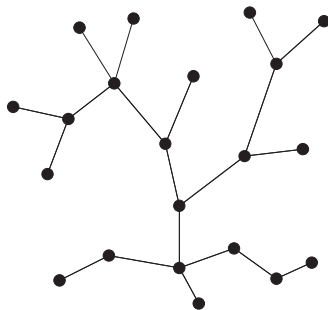


Fig. 1.5.1. A tree

[1.6.1]
[1.9.5]
[4.2.9]

Theorem 1.5.1. *The following assertions are equivalent for a graph T :*

- (i) T is a tree;
- (ii) Any two vertices of T are linked by a unique path in T ;
- (iii) T is minimally connected, i.e. T is connected but $T - e$ is disconnected for every edge $e \in T$;
- (iv) T is maximally acyclic, i.e. T contains no cycle but $T + xy$ does, for any two non-adjacent vertices $x, y \in T$. $\square$

xTy

The proof of Theorem 1.5.1 is straightforward, and a good exercise for anyone not yet familiar with all the notions it relates. Extending our notation for paths from Section 1.3, we write xTy for the unique path in a tree T between two vertices x, y (see (ii) above).

chord

A common application of Theorem 1.5.1 is that every connected graph contains a spanning tree: take a minimal connected spanning subgraph and use (iii). Figure 1.4.1 shows a spanning tree in each of the three components of the graph depicted. When T is a spanning tree of G , the edges in $E(G) \setminus E(T)$ are the *chords* of T in G .

[1.9.5]
[2.4.4]
[4.2.9]

Corollary 1.5.2. *A connected graph with n vertices is a tree if and only if it has $n - 1$ edges.*

Proof. For the forward implication, enumerate the vertices of a tree T as in Proposition 1.4.1. As T is acyclic, every vertex is adjacent to only one earlier vertex. Now $|T| = n - 1$ follows by induction on n .

Conversely, let G be any connected graph with n vertices and $n - 1$ edges. Let T be a spanning tree in G . Since T has $n - 1$ edges by the first implication, it follows that $T = G$. $\square$

[9.2.1]
[9.2.3]

Corollary 1.5.3. *If T is a tree and G is any graph with $\delta(G) \geq |T| - 1$, then T is (isomorphic to) a subgraph of G .*

Proof. Map the vertices of T to G inductively, following their enumeration from Proposition 1.4.1 applied to T . $\square$

Sometimes it is convenient to consider one vertex of a tree as special; such a vertex is then called the *root* of this tree. A tree T with a fixed root r is a *rooted tree*. Writing $x \leq y$ for $x \in rTy$ then defines a partial ordering on $V(T)$, the *tree-order* associated with T and r . We shall think of this ordering as expressing ‘height’: if $x < y$ we say that x lies *below* y in T , we call

$$[y] := \{x \mid x \leq y\} \quad \text{and} \quad [x] := \{y \mid y \geq x\}$$

the *down-closure* of y and the *up-closure* of x , and so on. A set $X \subseteq V(T)$ that equals its up-closure, i.e. which satisfies $X = [X] := \bigcup_{x \in X} [x]$, is *closed upwards*, or an *up-set* in T . Similarly, there are *down-closed* sets, or *down-sets* etc..

Note that the root of T is the least element in its tree-order, the leaves are its maximal elements, the ends of any edge of T are comparable, and the down-closure of every vertex is a *chain*, a set of pairwise comparable elements. (Proofs?) The vertices at distance k from the root have *height* k and form the k th *level* of T .

A rooted tree T contained in a graph G is called *normal* in G if the ends of every T -path in G are comparable in the tree-order of T . If T spans G , this amounts to requiring that two vertices of T must be comparable whenever they are adjacent in G ; see Figure 1.5.2.

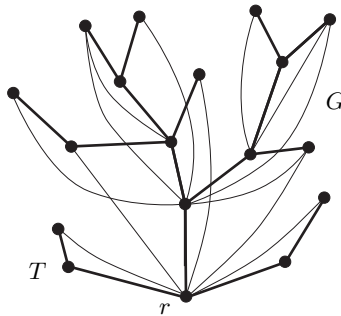


Fig. 1.5.2. A normal spanning tree with root r

A normal tree T in G can be a powerful tool for examining the structure of G , because G reflects the separation properties of T :

Lemma 1.5.4. *Let T be a normal tree in G .*

- (i) *Any two vertices x, y of T that are incomparable in its tree-order are separated in G by the set $[x] \cap [y]$.*
- (ii) *If $V(T) = V(G) =: V$ and $S \subseteq V$ is down-closed, then the components of $G - S$ are spanned by the sets $[x]$ with x minimal in $V \setminus S$.*

root

tree-order

up/above
down/below

$[t], [t]$

down-closure
up-closure

chain

height, level

normal tree

[8.2.3]
[8.6.8]

Proof. (i) As x and y are incomparable, neither of them lies in $[x] \cap [y]$. So it suffices to show that every x - y path P in G meets $[x] \cap [y]$. Let $t_1, \dots, t_n$ be a minimal sequence of vertices in $P \cap T$ such that $t_1 = x$ and $t_n = y$ and t_i and t_{i+1} are comparable in the tree-order of T for all i . (Such a sequence exists: the set of all vertices in $P \cap T$, in their natural order as they occur on P , has this property because T is normal and every segment $t_i P t_{i+1}$ is either an edge of T or a T -path.) In our minimal sequence we cannot have $t_{i-1} < t_i > t_{i+1}$ for any i , since t_{i-1} and t_{i+1} would then be comparable, and deleting t_i would yield a smaller such sequence. Thus, our sequence has the form

$$x = t_1 > \dots > t_k < \dots < t_n = y$$

for some $k \in \{1, \dots, n\}$ (even with $k \leq 3$, by the minimality of n). As $t_k \in [x] \cap [y] \cap V(P)$, our proof is complete.

(ii) Every set $[x]$ as in (ii) is connected in T , and hence in G . It lies in $V \setminus S$, because $x \notin S$ and S is down-closed. As every vertex in $V \setminus S$ lies above some minimal such vertex x , these sets $[x]$ have union $V \setminus S$.

For distinct x and x' , the connected sets $[x]$ and $[x']$ are disjoint, and not joined by an edge of G , because $[x] \cap [x'] \subseteq S$ separates x from x' in G , by (i). So the sets $[x]$ span maximal connected subgraphs, components, in $G - S$, and these are all its components. $\square$

Normal spanning trees are also called *depth-first search trees*, because of the way they arise in computer searches on graphs (Exercise 29). This fact is often used to prove their existence, which can also be shown by a very short and clever induction (Exercise 28). The following constructive proof, however, illuminates better how normal trees capture the structure of their host graphs.

[6.5.3]
[8.2.4]

Proposition 1.5.5. *Every connected graph has a normal spanning tree.*

Proof. Let G be a connected graph. Let T be any maximal normal tree in G ; we show that $V(T) = V(G)$.

Suppose not, and let C be a component of $G - T$. As T is normal, $N(C)$ is a chain in T . Let x be its greatest element, and let $y \in C$ be adjacent to x . Let T' be the tree obtained from T by joining y to x ; the tree-order of T' then extends that of T . We shall derive a contradiction by showing that T' is also normal in G .

Let P be a T' -path in G . If the ends of P both lie in T , then they are comparable in the tree-order of T (and hence in that of T'), because then P is also a T -path and T is normal in G by assumption. If not, then y is one end of P , so P lies in C except for its other end z , which lies in $N(C)$. Then $z \leq x$, by the choice of x . For our proof that y and z are comparable it thus suffices to show that $x < y$, i.e. that $x \in rT'y$. This, however, is clear since y is a leaf of T' with neighbour x . $\square$

1.6 Bipartite graphs

Let $r \geq 2$ be an integer. A graph $G = (V, E)$ is called *r-partite* if V admits a partition into r classes such that every edge has its ends in different classes: vertices in the same partition class must not be adjacent. Instead of ‘2-partite’ one usually says *bipartite*.

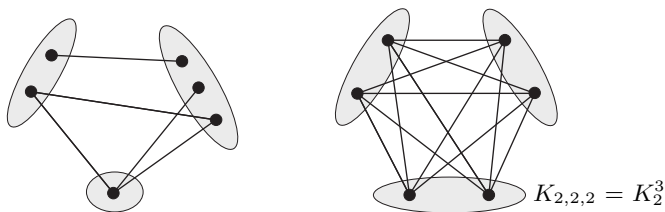
*r-partite**bipartite*

Fig. 1.6.1. Two 3-partite graphs

An r -partite graph in which every two vertices from different partition classes are adjacent is called *complete*; the complete r -partite graphs for all r together are the *complete multipartite* graphs. The complete r -partite graph $\overline{K^{n_1}} * \dots * \overline{K^{n_r}}$ is denoted by $K_{n_1, \dots, n_r}$; if $n_1 = \dots = n_r =: s$, we abbreviate this to K_s^r . Thus, K_s^r is the complete r -partite graph in which every partition class contains exactly s vertices.⁶ (Figure 1.6.1 shows the example of the octahedron K_2^3 ; compare its drawing with that in Figure 1.4.3.) Graphs of the form $K_{1,n}$ are called *stars*; the vertex in the singleton partition class of this $K_{1,n}$ is the star’s *centre*.

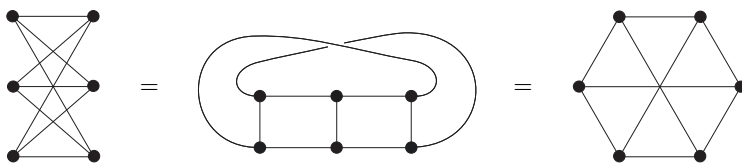
*complete
r-partite* $K_{n_1, \dots, n_r}$
 K_s^r *star
centre*

Fig. 1.6.2. Three drawings of the bipartite graph $K_{3,3} = K_3^2$

Clearly, a bipartite graph cannot contain an *odd cycle*, a cycle of odd length. In fact, the bipartite graphs are characterized by this property:

odd cycle

Proposition 1.6.1. *A graph is bipartite if and only if it contains no odd cycle.*

[1.9.4]
[5.3.1]
[6.4.2]

⁶ Note that we obtain a K_s^r if we replace each vertex of a K^r by an independent s -set; our notation of K_s^r is intended to hint at this connection.